MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER ON-LINE PROCTORED EXAMINATIONS

DECEMBER 2021

RENEWABLE ENERGY [ELE 4306] (OPEN ELECTIVE – IV)

REVISED CREDIT SYSTEM

Time: 75 Minutes + 10 Minutes **Date: 29 December 2021** **Max. Marks: 20**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Time: 75 minutes for writing + 10 minutes for uploading.

1A. Estimate the monthly average of the daily global radiation on a horizontal surface at Nagpur (21° 8' N, 79° 5' E) during the month of October 15 of a non-leap year, if the average sunshine hour per day is 7.3 hours. Assume a = 0.27 and b = 0.5. ***(04)***

1B. Explain the working principle of closed cycle Ocean thermal Energy conversion with a neat diagram. ***(03)***

1C. Find the angle subtended by beam radiation with the normal to a flat-plate collector at 9 a.m. for the day on November 3. The collector is in Delhi (28° 35' N, 77° 12' E), inclined at an angle of 36° with the horizontal and is facing due south. ***(03)***

2A. A Darrius rotor has the following diameters a=2.5m b=2m. as shown in Fig. Q. 2A. If it produces 3Kw of mechanical shaft power at velocity of 10m/s, calculate the power coefficient and also calculate the wind power.

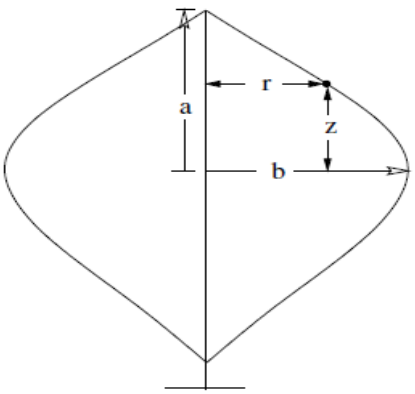


Fig. Q. 2A ***(03)***

2B. The wind power production company acquires a land 12X12 km land to install wind turbines of 50m blade length. Calculate the power production per land area when the wind power density at the hub is 350W/m^2, the coefficient of performance is 0.42 and the overall efficiency of turbine generator system is 93%. Assume the whole site the number of turbines are installed is 180 turbines, the array efficiency is 82%. ***(04)***

2C. With neat Schematic Explain a single process conventional digester. What are the main features of continuous plant. ***(03)***